# Learning to Distrust Misleading Text: Preference Optimization for Reducing Text Bias in Vision-Language Models

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## Abstract

*Vision-language models (VLMs) often treat textual context as a shortcut, even when that text conflicts with visual evidence. This "blind faith in text" is especially problematic in document and visual question answering, where auxiliary text may be incomplete, stale, adversarial, or automatically retrieved. We study whether preference optimization can reduce this failure mode without sacrificing normal visual question answering performance. The goal is not to teach the model to ignore all text: auxiliary text is often useful when it agrees with the image, and the model usually does not know in advance whether the text is reliable. Starting from corrupted-text evaluations, we mine model-specific text-following errors: cases where the model is wrong and its wrong answer is copied from the corrupted auxiliary text. We then train VLMs with Direct Preference Optimization (DPO), preferring the image-grounded answer over the model's own text-following error. On DocVQA with Qwen3-VL-8B, text-following span DPO improves corrupted accuracy from 0.715 to 0.915 and reduces the fraction of incorrect answers copied from corrupted text from 89.5% to 47.1%. A multi-condition error-mined extension further improves accuracy to 0.920 and reduces this text-following rate to 43.8%. By contrast, GRPO-style sampled-answer optimization and auxiliary SFT preservation do not improve over DPO. Experiments on VQAv2 and transfer to Qwen2-VL and LLaVA-Next show that the method works best when the base VLM already has sufficient visual and OCR competence. These results suggest that high-precision preference pairs are an effective and practical mechanism for correcting text bias in VLMs, while also revealing clear boundary conditions for the approach.*

## 1. Introduction

Vision-language models (VLMs) are increasingly used in settings where images are accompanied by textual context: document images with OCR, screenshots with metadata, retrieved descriptions, captions, or user-provided hints. Such text can be helpful when it is correct, but it can also become a shortcut. When textual context conflicts with the image, a VLM may answer from the text rather than from visual evidence. Deng et al. [3] describe this phenomenon as "blind faith in text", showing that several VLMs disproportionately trust textual information under conflict.

This failure is not merely a benchmark artifact. In practical systems, auxiliary text is often produced by retrieval, OCR, captioning, or prior model predictions. All of these sources can be noisy. If a VLM over-trusts such text, the model can produce confident but visually ungrounded answers. For document VQA, this is particularly important: the model must decide whether the image or the accompanying text should be trusted when they disagree.

A natural objection is that the model could simply ignore the auxiliary text whenever corruption is possible. This is not a satisfactory solution. First, in real use the system is not told whether the text is corrupted, matched, or irrelevant. Second, auxiliary text is often genuinely useful: when it matches the image, ignoring it can remove information that helps the model answer. Third, blind text removal does not teach conflict resolution; it only changes the input distribution. The desired behavior is conditional trust: use text when it is supported by the image, but reject it when it contradicts visual evidence.

We ask a targeted question: *Can we reduce text-following failures by directly training a VLM to prefer image-grounded answers over its own corrupted-text errors?* Our key observation is that the corrupted-text setting naturally produces preference pairs. If a model answers incorrectly and its prediction appears in the corrupted auxiliary text, then the model has likely followed the misleading text. The ground-truth answer can be used as the preferred response, and the model's text-following prediction can be used as the dispreferred response. This gives a high-precision preference signal without manual labeling.

We instantiate this idea with Direct Preference Optimization (DPO) [11]. Unlike online reinforcement learning methods such as PPO or GRPO [12], DPO directly opti-

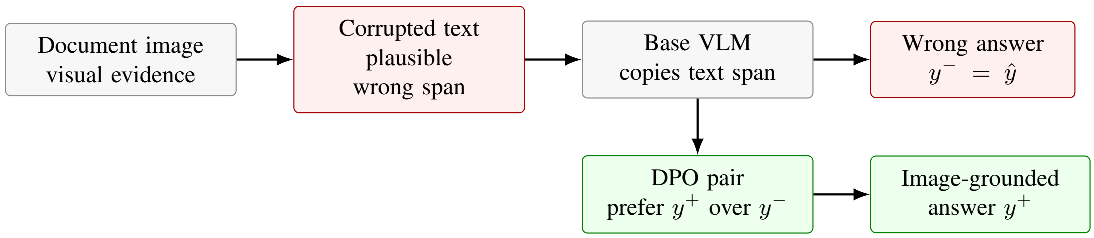


Figure 1. Overview of our mitigation. We mine cases where the base VLM follows corrupted auxiliary text, then train a DPO preference that ranks the image-grounded answer above the model's own text-following error.

mizes pairwise preferences. This is well aligned with our setting: the contrastive pair is already available as {image-grounded answer, text-following wrong answer}. Prior work already showed that supervised fine-tuning (SFT) with text augmentation can mitigate blind faith in text [3]. Our goal is not to deny that result, but to ask whether a more targeted preference signal can address the same failure mode. Instead of training on a broad mixture of text-only, original VQA, match, corruption, and irrelevance samples, we mine the specific cases where the model follows corrupted text and directly contrast that answer with the ground truth. We compare DPO against GRPO-style sampled-answer optimization, auxiliary SFT preservation, model transfer, and dataset transfer.

Figure 1 summarizes our mitigation. We treat Words or Vision as evidence that the conflict exists, and focus on which training signal most directly changes the behavior. Removing text changes the input rather than teaching conflict resolution; generic SFT supplies correct answers without explicitly identifying the misleading answer; online RL depends on candidate sampling and reward shaping. We therefore use preference pairs derived from actual text-following failures, where the chosen and rejected answers instantiate the desired behavioral reversal.

Our study makes three empirical findings. First, on DocVQA with Qwen3-VL-8B, DPO sharply improves corrupted-text robustness while reducing the rate of text-copying among remaining errors. Second, broadening DPO from corrupted-only text-following errors to multi-condition error-mined pairs yields the strongest result, suggesting that condition-specific error mining can improve stability. Third, the method has an important boundary condition: it works best when the base VLM already has enough visual/OCR competence. Qwen-family models benefit substantially, while LLaVA-Next models under our prompt and template show weak DocVQA base performance and limited benefit.

We summarize our contributions as follows:

- We define an automatic text-following error mining procedure for corrupted-text VQA, requiring both incorrectness and overlap between the wrong prediction and corrupted auxiliary text.
- We show that DPO on these mined pairs substantially reduces text bias on DocVQA and transfers directionally to VQAv2 and Qwen2-VL.
- We position DPO relative to the SFT mitigation studied by Deng et al. [3], and compare it with GRPO, auxiliary SFT preservation, multi-condition error mining, and model transfer.

## 2. Related Work

**Text bias in VLMs.** VLMs are trained to fuse visual and textual inputs, but fusion can become asymmetric when textual context is easier to exploit than visual evidence. The most directly related work is Words or Vision [3], which introduces corrupted textual variations and shows that VLMs may trust text over vision when the two conflict. Our work follows this diagnostic setting but focuses on mitigation through preference optimization. Rather than only measuring the degradation under corrupted text, we use the model's own corrupted-text failures as training signal. Deng et al. [3] also study SFT with text augmentation and show that increasing SFT data reduces reliance on corrupted or irrelevant text while preserving the ability to use matched text. They further note that text-only data helps prevent models from rejecting text indiscriminately. We build on this insight: the goal is conditional text use, not blanket text removal. Our contribution is to replace broad SFT augmentation with model-specific preference pairs mined from actual text-following failures. Other recent conflict-oriented evaluations similarly study how VLMs behave when image and text cues disagree. Mixed Signals [10] constructs mismatched image-text pairs across domains and studies model biases under conflicting signals. These works motivate our focus on conflict resolution rather than ordinary VQA accuracy alone. In contrast to benchmark-only studies, we explicitly turn conflict errors into training pairs.

**Document and visual question answering.** DocVQA [7] evaluates question answering over document images, requiring OCR, layout understanding, and reasoning over visual text. VQAv2 [4] evaluates open-ended visual question answering with reduced language priors compared to earlier VQA datasets. We use DocVQA as the main testbed because conflict between visual document evidence and auxiliary text is central to the task. VQAv2 is used as a transfer check to test whether the mitigation is specific to document images. Text-centric VQA datasets such as TextVQA [13], ST-VQA [2], OCR-VQA [9], and InfographicVQA [8] emphasize reading and reasoning over text inside images. These tasks are related but not identical to our corrupted-auxiliary-text setting. In TextVQA, scene text is often the correct visual evidence; in our DocVQA conflict setting, auxiliary text can be an external shortcut that contradicts the image. This distinction explains why we report TextVQA as a boundary check rather than as the main benchmark.

**Preference optimization.** Preference optimization has become a common alternative to supervised fine-tuning for aligning generative models. DPO [11] directly optimizes a policy from chosen/rejected pairs without training an explicit reward model or running online RL. This makes it attractive for our setting because corrupted-text evaluation yields natural preference pairs. GRPO [12] is an online reinforcement learning method that estimates a group-relative baseline from multiple sampled responses. We compare DPO with GRPO-style optimization and find that free-form sampled-answer rewards are less effective for this failure mode. Several recent works adapt preference optimization to multimodal models. mDPO [14] argues that standard multimodal DPO can suffer from unconditional preference learning, where the image condition is underused. Re-Align [16] constructs dual preference data with retrieval to incorporate visual preference signals. Task Preference Optimization [17] uses task-specific visual supervision to improve multimodal perception, and on-policy DPO has been explored for hallucination mitigation [18]. Our work is complementary: rather than building generic human or retrieval preferences, we mine preference pairs from a specific modality-conflict failure mode.

**Modern VLMs.** Qwen2-VL [15] and Qwen2.5-VL [1] emphasize dynamic-resolution visual processing and strong document/OCR capability. LLaVA-style models [5, 6] provide a widely used visual instruction tuning framework. Our transfer experiments suggest that these model families differ substantially in the base competence needed for text-bias correction.

# 3. Method

## 3.1. Problem Setup

Each example consists of an image $x$, a question $q$, auxiliary text $t$, and a ground-truth answer $y$. We consider three main textual conditions. In the *match* condition, the auxiliary text is consistent with the image. In the *irrelevant* condition, the text is unrelated. In the *corrupted* condition, the text contains a plausible but wrong answer. The desired behavior is not to ignore text unconditionally. Instead, the model should use text when it is reliable and fall back to visual evidence when text conflicts with the image.

## 3.2. Mining Text-Following Errors

We first evaluate a base VLM on corrupted examples. Let $\hat{y}$ be the model prediction. We define a *text-following error* when all of the following hold:

1. the model prediction is incorrect;
2. the prediction is non-empty and does not match the ground truth;
3. the prediction appears in the corrupted auxiliary text.

For the third condition, we use normalized string matching. For multi-token answers, we also accept high token overlap with the auxiliary text. This filtering is important: an arbitrary wrong answer is not necessarily a text-bias error, but an incorrect prediction copied from corrupted text is a much stronger signal.

For each mined error, we construct a DPO pair:

$$y^+ = y, \qquad y^- = \hat{y}, \tag{1}$$

where $y^+$ is the image-grounded answer and $y^-$ is the model's text-following wrong answer. We call this dataset *text-following span DPO*.

**Why error mining matters.** The filtering step is not merely a data-cleaning heuristic. It determines what behavioral claim the training data supports. If we trained on every wrong prediction, the rejected answer could correspond to many different failure modes: OCR failure, answer-format mismatch, arithmetic error, hallucination, or missing document evidence. Such pairs would still be valid supervised corrections, but they would not specifically target blind faith in text. By requiring the wrong answer to appear in the corrupted auxiliary text, we isolate a narrower intervention: the model saw a misleading textual cue and produced that cue as its answer. This makes the preference pair interpretable as a conflict-resolution example rather than as generic answer correction.

This distinction is important for our comparison with SFT and GRPO: SFT teaches the correct answer, GRPO must discover useful alternatives through sampling, while our mined DPO pair directly asks the model to reverse a concrete preference it already exhibited under conflict.

## 3.3. Why This Is Not Text Removal

A natural baseline is to delete the auxiliary text or instruct the model to ignore it. However, this changes the task rather than solving the conflict. In real document and visual-question-answering settings, auxiliary text may come from OCR, retrieved context, metadata, or captions. Some of it is useful, some is irrelevant, and some may be wrong. The model must therefore decide whether the text is supported by the image, not merely whether text is present.

Our training data encodes this conditional behavior through the surrounding conditions: corrupted examples provide the main negative signal, match examples test reliable text use, and irrelevant/no-text examples test whether the model can answer without overusing unsupported context. Thus the desired policy is not

$$\text{always distrust text,} \tag{2}$$

but rather

$$\text{trust text only when it is visually grounded.} \tag{3}$$

This is why preservation metrics are central in our evaluation. An approach that improves corrupted accuracy by discarding all auxiliary text would be a weaker solution than one that improves corrupted robustness while preserving match and irrelevant performance.

### 3.4. DPO Objective

Given a policy $\pi_\theta$ and a reference policy $\pi_{\text{ref}}$, DPO optimizes the preference that $y^+$ should be more likely than $y^-$ under the prompt $(x, q, t)$:

$$\begin{aligned}\mathcal{L}_{\text{DPO}} = -\log \sigma\Big(\beta\big[\log \pi_\theta(y^+|x,q,t) - \log \pi_{\text{ref}}(y^+|x,q,t) \\ - \log \pi_\theta(y^-|x,q,t) + \log \pi_{\text{ref}}(y^-|x,q,t)\big]\Big).\end{aligned} \tag{4}$$

In our setting, this objective directly encodes the desired correction: the image-grounded answer should be preferred over the corrupted-text answer.

### 3.5. Multi-Condition Error-Mined DPO

Text-following span DPO uses only corrupted examples. To reduce drift and improve stability, we also test a multi-condition extension. We mine errors from corrupted, match, irrelevant, and no-text evaluations. For corrupted examples, we keep the strict text-following filter: the wrong prediction must appear in corrupted auxiliary text. For match, irrelevant, and no-text examples, we use a broader condition-specific error filter: the prediction must be incorrect, non-empty, and distinct from the ground truth.

We assign lower weights to non-corrupted errors:

$$\begin{aligned} w_{\text{corrupted}} = 1.0, \quad w_{\text{match}} = 0.25, \\ w_{\text{irrelevant}} = 0.5, \quad w_{\text{no-text}} = 0.5. \end{aligned} \tag{5}$$

This preserves the central text-bias signal while adding weak regularization from other conditions.

### 3.6. GRPO-Style Baselines

We also test GRPO-style optimization. For each prompt, the model samples multiple candidate answers. A reward function gives positive reward for matching the chosen answer and penalties for matching the rejected answer, misleading spans mined from corrupted text, or any answer copied from the auxiliary text. The reward also mildly encourages short answer-like completions. This tests whether online sampled-answer optimization can discover the same correction signal without explicit pairwise DPO.

## 4. Experiments

### 4.1. Metrics

We report three complementary metrics. *Corrupted accuracy* measures task performance when auxiliary text is misleading. *Incorrect aux-hit* measures the fraction of incorrect predictions that appear in the corrupted auxiliary text. Lower incorrect aux-hit indicates less text-following among remaining errors, but it is not sufficient by itself: a model could reduce aux-hit by hallucinating unrelated answers. Therefore, we interpret it together with corrupted accuracy and preservation on match/irrelevant conditions. We also report corrected/regressed counts relative to the base model when available.

### 4.2. Implementation Details

Unless otherwise stated, we train Qwen3-VL-8B-Instruct with LoRA adapters. The main DocVQA split uses 800 corrupted examples for mining and a disjoint held-out corrupted set of 200 examples for evaluation. The text-following span dataset is built only from base-model failures satisfying the corrupted-text overlap filter. For the initial span-DPO run, we use LoRA rank 8, LoRA alpha 16, learning rate $10^{-5}$, DPO $\beta = 0.1$, and 700 DPO steps. The stronger rank-16 runs use LoRA alpha 32 with the same learning rate and number of steps. The multi-condition error-mined run uses 273 total preference rows, split into 245 train and 28 test rows, with source counts of 200 corrupted, 7 match, 33 irrelevant, and 33 no-text errors. It trains for 900 DPO steps with LoRA rank 16, alpha 32, learning rate $10^{-5}$, and $\beta = 0.1$.

All reported transfer experiments use the same evaluation code and the same model family-specific prompt template used during mining. This matters because our LLaVA-Next experiments show strong sensitivity to task and template compatibility. We therefore interpret LLaVA results as boundary evidence rather than as a claim that the method is model-agnostic.

**Corruption construction.** For DocVQA, corrupted auxiliary text is created by replacing the answer-bearing textual

Table 1. DocVQA corrupted held-out results with Qwen3-VL-8B.

| Method | Acc. | Incorrect aux-hit | Corr./Reg. |
|---|---|---|---|
| Base | 0.715 | 89.5% | – |
| TF-span DPO, r8 | 0.900 | 55.0% | 37 / 0 |
| TF-span DPO, r16 | 0.915 | 47.1% | 40 / 0 |
| Multi-cond. error DPO | **0.920** | **43.8%** | 41 / 0 |

hint with a plausible but incorrect answer span. The corrupted span is intentionally answer-like rather than random noise. This makes the setting difficult: a model that relies on surface text can produce a fluent and plausible answer, while a model that checks the image must reject the hint. For match and irrelevant conditions, the auxiliary text is either consistent with the ground-truth answer or unrelated to the queried evidence. Together, these conditions separate three behaviors that would be conflated in ordinary VQA accuracy: using helpful text, ignoring irrelevant text, and correcting misleading text.

We use exact normalized answer matching for VQA accuracy. For incorrect aux-hit, we measure whether an incorrect prediction overlaps the corrupted auxiliary answer span after normalization. Because aux-hit is computed only among wrong predictions, it diagnoses residual failure type rather than replacing accuracy.

### 4.3. DocVQA Main Results

DocVQA is our main testbed. We use Qwen3-VL-8B-Instruct as the primary model and evaluate on a held-out corrupted split. Table 1 shows that text-following span DPO substantially improves both accuracy and text-bias behavior. Increasing LoRA rank from 8 to 16 improves the result further, while the multi-condition error-mined variant gives the best overall performance.

The multi-condition result suggests that the highest-performing variant is not corrupted-only. However, the corrupted filter remains essential: corrupted rows still require the prediction to be copied from corrupted auxiliary text. The additional match, irrelevant, and no-text errors act as weak condition-specific regularizers rather than replacing the text-following signal.

The main DocVQA result is encouraging because it increases corrupted accuracy and lowers aux-hit without large match/irrelevant degradation.

### 4.4. Hyperparameter and Construction Ablations

We run a small $2 \times 2$ grid over DPO $\beta$ and LoRA rank on the same DocVQA split. Table 2 shows that increasing LoRA rank from 8 to 16 is more useful than increasing $\beta$ from 0.1 to 0.2. Rank 16 improves corrupted accuracy from 0.900 to 0.915 and reduces incorrect aux-hit from 55.0% to 47.1%, while changing $\beta$ does not change the corrupted metric in this grid.

Table 2. DocVQA DPO hyperparameter grid. Higher LoRA capacity matters more than $\beta$ in this setting.

| $\beta$ | Rank | Acc. | Aux-hit | Corr./Reg. | Match/Irr. |
|---|---|---|---|---|---|
| 0.1 | 8 | 0.900 | 55.0% | 37 / 0 | 1.00 / 0.96 |
| 0.2 | 8 | 0.900 | 55.0% | 37 / 0 | 0.98 / 0.91 |
| 0.1 | 16 | **0.915** | **47.1%** | 40 / 0 | 0.98 / 0.91 |
| 0.2 | 16 | **0.915** | **47.1%** | 40 / 0 | 0.98 / 0.91 |

Table 3. GRPO-style optimization does not improve over DPO on DocVQA.

| Variant | Init | Acc. | Incorrect aux-hit |
|---|---|---|---|
| Base | – | 0.715 | 89.5% |
| GRPO-only, 100 steps | base | 0.710 | 89.7% |
| GRPO-only, 300 steps | base | 0.705 | 89.8% |
| GRPO-only, dense reward | base | 0.705 | 89.8% |
| TF-span DPO | base | **0.900** | **55.0%** |
| DPO $\rightarrow$ GRPO | DPO | **0.900** | 60.0% |

Prompt-augmented counterfactual DPO, which reuses the same rejected answer across corrupted, no-text, match, and irrelevant wrappers, ties plain rank-16 DPO at 0.915 accuracy and 47.1% aux-hit. Multi-condition error-mined DPO instead uses the actual failed prediction from each condition as the rejected answer, improving to 0.920 accuracy and 43.8% aux-hit.

### 4.5. DPO versus GRPO

We compare DPO with GRPO-only and DPO-to-GRPO variants on the same disjoint DocVQA split. GRPO samples multiple answers per prompt and optimizes a reward that encourages correctness and penalizes copying corrupted text. Despite this denser reward, GRPO does not move the held-out greedy behavior meaningfully. Table 3 shows that DPO is the effective intervention in our setting.

DPO-to-GRPO changes only a few held-out predictions relative to DPO and does not improve accuracy. This supports our hypothesis: when high-precision chosen/rejected pairs are available, direct pairwise optimization provides a clearer learning signal than free-form sampled-answer rewards.

### 4.6. SFT: Prior Mitigation and Our Ablation

The original Words or Vision study already evaluates SFT with text augmentation as a mitigation strategy [3]. That SFT setting uses a broader training mixture, including original VQA, text-only, match, corruption, and irrelevance samples, and shows that SFT can reduce text reliance while maintaining the ability to benefit from matched text. This is an important baseline and motivates our own mitigation study.

Table 4. VQAv2 transfer results with Qwen3-VL-8B.

| Variant | Corr. Acc. | Aux-hit | Match | Irr. |
|---|---|---|---|---|
| Base | 0.545 | 65.9% | 0.92 | 0.77 |
| Span DPO | **0.645** | **40.8%** | 0.90 | 0.75 |
| Span DPO + SFT | 0.635 | 42.5% | 0.90 | 0.75 |

Our experiment is different. We do not reproduce the full augmented SFT recipe from Deng et al. [3]. Instead, we test a narrower question: after constructing text-following DPO pairs, can an auxiliary SFT preservation loss on match/irrelevant examples further stabilize the model? This is a preservation ablation, not a claim that SFT in general is ineffective. SFT is plausible because it directly teaches the model to output correct answers. However, the central error in our setting is not simply that the model does not know the answer; it is that the model prefers a misleading textual answer over visual evidence. SFT provides only a positive target, while DPO explicitly contrasts the image-grounded answer against the text-following wrong answer.

Our SFT experiment uses auxiliary preservation data from match and irrelevant conditions and mixes it with span DPO. As shown in Table 4, this does not improve over pure DPO on VQAv2: corrupted accuracy decreases from 0.645 to 0.635, incorrect aux-hit increases from 40.8% to 42.5%, and match/irrelevant preservation remains unchanged. Thus, in our experiments, auxiliary SFT preservation acts as a negative ablation, while the prior full SFT augmentation result remains a valid mitigation baseline from the original work.

### 4.7. VQAv2 Transfer Check

We repeat the disjoint split protocol on VQAv2. The effect is positive but weaker than on DocVQA. As shown in Table 4, span DPO improves corrupted accuracy by 10 points and reduces incorrect aux-hit, but match and irrelevant preservation drop slightly. Adding auxiliary SFT preservation does not recover this drop and slightly worsens corrupted performance.

### 4.8. TextVQA Boundary Check

We also test whether a DocVQA-trained span DPO adapter transfers to TextVQA. TextVQA is useful because it is text-centric, but it differs from DocVQA conflict: scene text is often the correct visual evidence rather than an external auxiliary shortcut. We therefore construct several TextVQA corruption variants, including same-image OCR distractors, cross-sample answers, and question-type matched answer hints.

Table 5 shows the strongest TextVQA conflict variant, question-type matched answer hints, on 200 examples. For Qwen3-VL, the base model already has low incorrect aux-hit under corrupted TextVQA hints, and DocVQA DPO gives only a small gain. Qwen2-VL shows a similar but slightly stronger positive direction. LLaVA-Next behaves differently: the conflict is stronger, but DocVQA DPO does not reliably fix it and can worsen corrupted accuracy.

Table 5. TextVQA question-type matched answer-hint transfer. DocVQA DPO transfers weakly to Qwen models but not reliably to LLaVA-Next.

| Model | Base Acc. | DPO Acc. | Base aux-hit | DPO aux-hit |
|---|---|---|---|---|
| Qwen3-VL-8B | 0.821 | 0.827 | 11.5% | 8.3% |
| Qwen2-VL-7B | 0.832 | 0.845 | 23.1% | 8.3% |
| LLaVA-N-7B | 0.535 | 0.443 | 50.0% | 62.0% |
| LLaVA-N-13B | 0.612 | 0.602 | 35.7% | 28.8% |

Table 6. Model transfer on DocVQA. DPO helps Qwen2-VL but not LLaVA-Next under our setting.

| Model | Setting | Acc. | Aux-hit | Corr./Reg. |
|---|---|---|---|---|
| Qwen2-VL-7B | base | 0.570 | 93.0% | – |
| Qwen2-VL-7B | span DPO | **0.775** | 86.7% | 41 / 0 |
| LLaVA-N-7B | base | 0.105 | 90.5% | – |
| LLaVA-N-7B | span DPO | 0.100 | 91.7% | 1 / 2 |
| LLaVA-N-13B | base | 0.115 | 92.7% | – |
| LLaVA-N-13B | span DPO | 0.130 | 95.4% | 3 / 0 |

These results sharpen the scope of our claim. The method is not a universal accuracy booster for all text-heavy VQA tasks. It is most effective when the base model actually exhibits text-following failures under corrupted auxiliary text and already has enough visual/OCR competence to answer from the image.

### 4.9. Model Transfer

We test whether the same mitigation extends beyond Qwen3-VL. Table 6 summarizes Qwen2-VL-7B and LLaVA-Next results on DocVQA. Qwen2-VL benefits substantially, improving from 0.570 to 0.775 corrupted accuracy. LLaVA-Next, however, has very low base DocVQA accuracy under our prompt/template, and DPO provides little benefit.

These results show that the method is not simply a universal post-hoc fix. It requires a base model that can already solve the task visually; DPO then teaches the model to resist misleading text.

## 5. Analysis and Discussion

### 5.1. Why DPO Fits This Problem

The corrupted-text setting gives unusually clean preference pairs. For many failures, the model's wrong answer is directly copied from the corrupted auxiliary text. Thus the comparison is explicit: the image-grounded answer should be preferred over the text-following wrong answer. DPO optimizes exactly this comparison. This complements, rather than invalidates, the SFT mitigation in Words or Vision [3]. SFT with text augmentation teaches the model from a balanced mixture of cases and can reduce broad text reliance. Our DPO variant instead uses a narrower but sharper signal: for each mined failure, it tells the model which specific text-derived answer should lose to the visual answer. In contrast, GRPO must first sample useful candidate answers and then rely on a hand-designed reward. If the sampled candidates rarely contain the correct answer, or if most candidates are similar text-following errors, the relative reward signal is weak. This explains why GRPO logs can show diagnostic reward events while greedy held-out predictions barely change.

Our DPO construction uses the benchmark answer as the chosen response. This is appropriate because the task is not open-ended preference alignment but targeted VQA correction: the correct answer should beat an incorrect answer copied from misleading text. The advantage over SFT is contrastive. SFT says only what to output, while DPO also identifies the tempting text-derived answer that should be suppressed.

## 5.2. Why Not Simply Ignore the Text?

Ignoring auxiliary text is an appealing but incomplete baseline. It would avoid some corrupted-text failures, but it does not solve the actual decision problem faced by a VLM. At inference time, the model is not given a label saying whether the text is matched, irrelevant, or corrupted. If the model learns to discard text wholesale, it can lose useful information in match cases and may fail to exploit reliable OCR or retrieved context.

Our objective is therefore not "use less text" but "use text conditionally". The model should keep using textual context when it is visually supported and reject it when it conflicts with the image. DPO is appropriate because it changes the model's preference in exactly the conflict cases we can identify:

$$\text{image-grounded answer} \succ \text{text-following wrong answer}. \tag{6}$$

This is different from input ablation or prompt-level instruction to ignore text. The latter can reduce exposure to misleading text, but it does not teach the model which answer should be preferred when both visual evidence and misleading text are present. Our use of match and irrelevant preservation metrics is intended to check precisely this point: a successful method should improve corrupted robustness without collapsing into text avoidance.

## 5.3. What Counts as Solving Text Bias?

We therefore define a successful correction operationally through four criteria: held-out corrupted accuracy should improve, incorrect aux-hit should decrease, match/irrelevant settings should be preserved, and the trend should be at least partially reproducible across nearby settings such as DPO hyperparameters or related VQA benchmarks. Our DocVQA results satisfy the first three criteria strongly and the fourth criterion partially: Qwen-family transfer is positive, while LLaVA and TextVQA reveal clear boundary conditions.

## 5.4. Why Multi-Condition Error Mining Helps

Text-following span DPO is the core filtering strategy, but the best result comes from multi-condition error-mined DPO. This does not mean that unfiltered errors are sufficient. Rather, corrupted errors remain strictly filtered for text-following behavior, while other conditions contribute lower-weight stability signals. This helps prevent the model from over-specializing to the corrupted prompt format. The gain is modest because non-corrupted failures are sparse in DocVQA, but it consistently improves the main held-out metric.

## 5.5. Boundary Conditions

Our transfer experiments reveal a clear boundary condition. The method works when the base VLM already has sufficient visual and OCR competence. Qwen-family models satisfy this condition on DocVQA and improve substantially. LLaVA-Next models, under our prompt/template, have very low base DocVQA accuracy. In this regime, DPO cannot create document-reading ability from scratch; it can only alter preferences among answers the model can already represent.

TextVQA experiments lead to a related conclusion. DocVQA-trained DPO transfers weakly to TextVQA and does not reliably fix LLaVA-specific TextVQA conflict failures. Even after mining larger LLaVA-specific TextVQA DPO data and adding strict+preserve pairs, corrupted performance does not improve. This suggests that TextVQA conflict involves different OCR and answer-format dynamics than DocVQA. Therefore, we treat TextVQA and LLaVA results as boundary analyses rather than main success claims.

## 5.6. Limitations

Our study has several limitations. First, the text-following filter is operational rather than causal; accidental overlap with corrupted text is possible. Second, most experiments use one held-out split and limited seeds due to compute constraints. Third, our GRPO implementation uses free-form sampled answers, and candidate-constrained variants may be stronger. Finally, the method assumes ground-truth answers for preference construction, making it suitable for supervised mitigation but not directly for fully unlabeled deployment.

## 6. Conclusion

We studied preference optimization as a mitigation for VLM text bias under corrupted auxiliary text. By mining text-following errors and training with DPO, we convert a diagnostic failure mode into a direct preference-learning signal. On DocVQA, DPO improves corrupted-text accuracy and reduces copying from misleading text.

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